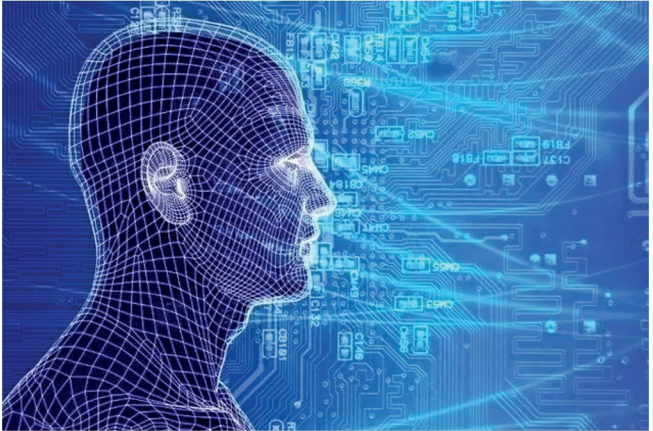


Exploring Front-end Computer Vision

Computer vision tasks include methods for acquiring, processing, analysing and understanding digital images, and in general, deal with the extraction of multi-dimensional data from the real world in order to produce numerical or symbolic information.



Computer vision (CV) is a discipline that relates to the visual perception and analysis of computers and cameras. The visual input method for computers is a camera. A majority of all computer vision algorithms focus on extrapolating interesting features from images/videos that are captured by a camera. This field has many applications in the field of robotics. For example, the preliminary versions of Stanford University's Stanley (a self-driving car) used a pair of stereo cameras for visual perception.

Technology today is shifting to a more cloud and Internet oriented setting. Traditional software is being replaced by Web apps. If eventually, everything is going to be ported to a Web platform, it would be wise to start incorporating the Web into upcoming technologies. Similarly, one could think of shifting CV to a browser platform as well. In fact, there are various libraries that provide browser based support for

computer vision. These include *Tracking.js*. First, let it be clear that a browser based system for this article refers to front-end code only, involving just HTML5, CSS, and JavaScript.

Basic computer vision and the browser

Computations are carried out upon images, with the fundamental unit being a pixel. Algorithms involve mathematical operations on a pixel or a group of pixels. This article addresses a few hackneyed CV algorithms and their ports to a front-end system. To start with, basic concepts like images and canvas are to be understood first.

An HTML image element refers to the '`<img></img>`' tag. It is, essentially, adding an image to a Web page. Similarly, to process or display any graphical units, the '`<canvas></canvas>`' element is used. Each of these elements has attributes such as height, width, etc, and is referred to via an ID. The